

SUNIX Linux Kernel Module

SDC Expansion Board Driver

Driver Patch Update

Driver Ver. 2.1.1.0

Ver. 1.0

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1. Issue description:

Sometimes, Even though the Linux kernel is within the scope supported by SDC, a compilation error occurs due to differences in header file definitions from Kernel.org, as seen in RHEL 9.4 (Kernel 5.14.0-427).

```
[root@localhost driver]# make
Compile SDC driver. (kernel 5.14.0-427)

make CFLAGS_MODULE+=-DNETCAN CFLAGS_MODULE+=-Wframe-larger-than=2048 EXTRA_CFLAGS+=-DSDC_CREATE_UART=1 -DSDC_CREATE_DIO=1 -DSDC_CREATE_SPI=1 -DSDC_CREATE_CAN=1 -DSDC_CREATE_PARPORT=0" -C /lib/modules/5.14.0-427.42.1.el9_4.x86_64/build M=
/home/test/SDCLinuxExpansionBoardDriver_ /driver modules
make[1]: Entering directory '/usr/src/kernels/5.14.0-427.42.1.el9_4.x86_64'
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.o
In file included from ./include/linux/linkage.h:7,
                 from ./include/linux/kernel.h:8,
                 from /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.c:9:
./include/linux/export.h:117:22: error: passing argument 1 of 'class_create' from incompatible pointer type [-Werror=incompatible-pointer-types]
   17 | #define THIS_MODULE (&__this_module)
      |                      ^
/home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.c:696:34: note: in expansion of macro 'THIS_MODULE'
   696 |     dev_class = class_create(THIS_MODULE, DRV_NAME);
      |                               ^
In file included from ./include/linux/device.h:31,
                 from ./include/linux/pm_runtime.h:11,
                 from /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.c:11:
./include/linux/device/class.h:230:54: note: expected 'const char *' but argument is of type 'struct module *'
   230 | struct class * __must_check class_create(const char *name);
      |                                     ^
/home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.c:696:21: error: too many arguments to function 'class_create'
   696 |     dev_class = class_create(THIS_MODULE, DRV_NAME);
      |                       ^
In file included from ./include/linux/device.h:31,
                 from ./include/linux/pm_runtime.h:11,
                 from /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.c:11:
./include/linux/device/class.h:230:29: note: declared here
   230 | struct class * __must_check class_create(const char *name);
      |                               ^
/home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.c:703:31: error: assignment to 'int (*)(const struct device *, struct kobj_uevent_env *)' from incompatible pointer type 'int (*)(struct device *, struct kobj_uevent_env *)' [-Werror=incompatible-pointer-types]
   703 |     dev_class->dev_uevent = sdc_config_uevent;
      |                               ^
cc1: some warnings being treated as errors
make[2]: *** [scripts/Makefile.build:299: /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.o] Error 1
make[1]: *** [Makefile:1936: /home/test/SDCLinuxExpansionBoardDriver_ /driver] Error 2
make[1]: Leaving directory '/usr/src/kernels/5.14.0-427.42.1.el9_4.x86_64'
make: *** [Makefile:70: default] Error 2
```

2. Driver patch update

This document uses the RHEL 9.4 patch as an example to explain the operational steps.

```
[root@localhost SDCLinuxExpansionBoardDriver_ ]# ls patch/
RHEL_9.4.patch
```

1. Enter “./SDCLinuxExpansionBoardDriver_VX.X.X.X/driver” directory.
Enter command:

```
$patch -p1 < ../patch/RHEL_9.4.patch
```

```
[root@localhost SDCLinuxExpansionBoardDriver_ ]# cd driver/
[root@localhost driver]# patch -p1 < ../patch/RHEL_9.4.patch
patching file mcp251x/mcp251x.c
patching file mcp251xfd/mcp251xfd-core.c
patching file sdccanfd/sdc_canfd.c
patching file sdc_config.c
patching file sdc_dio.c
[root@localhost driver]#
```

2. Use “make” and “make install” to build and install driver.

\$make

```
[root@localhost driver]# make
Compile SDC driver. (kernel 5.14.0-427)
make CFLAGS_MODULE=-DNETCAN CFLAGS_MODULE=-Wframe-larger-than-2048 EXTRA_CFLAGS=-DSDCC_CREATE_UART=1 -DSDCC_CREATE_DIO=1 -DSDCC_CREATE_SPI=1 -DSDCC_CREATE_CAN=1 -DSDCC_CREATE_PARPORT=0 -C /lib/modules/5.14.0-427.42.1.el9_4.x86_64/build M=
/home/test/SDCLinuxExpansionBoardDriver_ /driver/modules
make[1]: Entering directory /usr/src/kernels/5.14.0-427.42.1.el9_4.x86_64
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/8250_sdc.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_dio.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/spi-sdc.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/mcp251x/mcp251x.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251x.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/mcp251xfd/mcp251xfd-core.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/mcp251xfd/mcp251xfd-crc16.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/mcp251xfd/mcp251xfd-regmap.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251xfd.o
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdcanfd/sdc_canfd.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/can_sdc.o
MODPOST /home/test/SDCLinuxExpansionBoardDriver_ /driver/Module.symvers
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/8250_sdc.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/8250_sdc.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/8250_sdc.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/8250_sdc.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/can_sdc.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/can_sdc.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/can_sdc.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/can_sdc.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_config.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_dio.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_dio.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_dio.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_dio.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251x.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251x.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251x.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251x.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251xfd.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251xfd.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251xfd.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/sdc_mcp251xfd.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/spi-sdc.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/spi-sdc.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/spi-sdc.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/spi-sdc.ko due to unavailability of vmlinux
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.mod.o
LD [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.ko
BTF [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.ko
Skipping BTF generation for /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.ko due to unavailability of vmlinux
make[1]: Leaving directory /usr/src/kernels/5.14.0-427.42.1.el9_4.x86_64
```

\$make install

```
[root@localhost driver]# make install
Install SDC driver. (kernel 5.14.0-427)
sudo cp -f ./obj/sunix-sdc.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/mfd
sudo cp -f ./obj/sdc_config.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/char
sudo cp -f ./obj/8250_sdc.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/tty/serial/8250
sudo cp -f ./obj/sdc_dio.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/char
sudo cp -f ./obj/spi-sdc.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/spi
sudo cp -f ./obj/sdc_mcp251x.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/net/can/spi
sudo cp -f ./obj/sdc_mcp251xfd.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/net/can/spi
sudo cp -f ./obj/can_sdc.ko /lib/modules/5.14.0-427.42.1.el9_4.x86_64/kernel/drivers/net/can
sudo depmod -a
[root@localhost driver]#
```